



VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST – APRIL 2024

22

COURSE: BCA
SUBJECT: JAVA Programming
PAPER CODE: BCA 202

SEMESTER: IV
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B

PART – A

10 Marks

Q. No.	CO	Question	Marks
1.	CO3	What is <i>name space collision</i> in Java? How does Java handle this situation?	3
2	CO1	Discuss the requirement of all the literals in the following Java program statement: - “public static void main (String args [])”	3
3	CO1	Explain any 4 inbuilt functions of String class in Java. Explain the difference between ‘==’ operator and ‘equals()’ method.	4

PART – B Attempt any 2 questions from part B

2*10 = 20 Marks

Q. No.	CO	Question	Marks
4	CO2	A. What is the difference between <i>abstract class</i> and <i>interface</i> ? B. What is <i>super</i> Keyword? Write programs to show the usage of all cases of <i>super</i> keyword.	10
5	CO2	A. What is the <i>order of constructor</i> calling in inheritance? Explain with the help of a program. B. Explain the concept of <i>run time polymorphism</i> with the help of a program.	10
6	CO4	A. What is the mechanism of <i>exception handling</i> in Java? B. Differentiate between <i>throw</i> and <i>throws</i> keyword. C. Write a program to show how Java implements custom exception (user defined exception).	10



VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST - APRIL 2024

COURSE: BCA
SUBJECT: Introduction to Data Science
PAPER CODE: BCA 212

SEMESTER: IV
M.MARKS: 30
TIME: 1 Hour 30 minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

10 Marks

PART - A

Attempt any FIVE of the following:

Q. No.	CO	Question	Marks
1.	1	How to make a NumPy array read-only?	2
2.	1	Differentiate between inferential and descriptive analysis	2
3.	3	Differentiate between loc() and iloc() in Pandas DataFrame.	2
4.	3	What is a series and how is it different from 1-D array, list and a Dictionary	2
5.	2	What is meant by dispersion? If the standard deviation of a given dataset is equal to zero, what can we say about the data values included in the given dataset?	2
6.	3	What are four different ways of creating arrays?	2
7.	2	The head chef of a flying Taco has received two dozen tomatoes from her supplier but she isn't ready to accept them. She knows from the invoice that the average weight of a tomato is 7.5 ounce, but she insists that all be of uniform weight. She will accept them only if the average weight is 7.5 ounce and the standard deviation be less than 0.5 ounce. Here the weights of the tomatoes 6.3 • 7.2 7.3 • 8.1 • 7.8 • 6.8 • 7.5 • 7.2 • 7.5 • 8.1 8.0 • 7.4 7.6 7.7 7.6 7.4 • 7.5 • 8.4 • 7.4 8.2 • 7.6 6.2 What is the chef's decision and why?	2

PART - B

2*10 = 20 Marks

PART - B

2-10 = 20 Marks

Q. No.	CO	Question	Marks																								
1.	1	i. Explain roles and responsibilities of Data Science professionals	5																								
	1	ii. Discuss different stages of a data science project	5																								
2.	1	i. Discuss in details major tasks of data preprocessing. In real world tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem.	5																								
	3, 4	ii. Given a data frame with rainfall data (day of week and rainfall inches). Write a code to find minimum, maximum, Q1, Q2, Q3, mean and standard deviation.	5																								
3.	3, 4	Create a data frame to store data for 10 students <table> <tr> <th>Name</th> <th>Age</th> <th>Semester -I marks out of 500</th> <th>Semester-II marks out of 500</th> <th>Attendance</th> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> a. Display details of the students who score more than 450 marks in semester I b. Display details of the students who score less than 250 marks in semester II c. Display details of the students who score minimum marks in semester II and semester I d. Display details of the students who have more than 75% attendance e. Insert two new records in data frame f. Add a new column for percentage of marks in both semester g. Add a new column corresponding to grades <table> <tr> <th>Both semester percentage</th> <th>Grade</th> </tr> <tr> <td>>=90</td> <td>O</td> </tr> <tr> <td>>=75 to <90</td> <td>A+</td> </tr> <tr> <td>>=60 to <75</td> <td>A</td> </tr> <tr> <td>>=50 to <60</td> <td>B+</td> </tr> <tr> <td>>=40 to <50</td> <td>B</td> </tr> <tr> <td><40</td> <td>F</td> </tr> </table>	Name	Age	Semester -I marks out of 500	Semester-II marks out of 500	Attendance						Both semester percentage	Grade	>=90	O	>=75 to <90	A+	>=60 to <75	A	>=50 to <60	B+	>=40 to <50	B	<40	F	10
Name	Age	Semester -I marks out of 500	Semester-II marks out of 500	Attendance																							
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<40	F																										
4.	2,4	Cost accountants often estimate overhead based on the level of production. At the Standard Knitting Co., they have collected information on overhead expenses and units produced at different plants, and want to estimate a regression equation to predict future overhead. <table> <tr> <th>Overheads</th> <td>191</td> <td>170</td> <td>272</td> <td>155</td> <td>280</td> <td>173</td> <td>234</td> <td>116</td> <td>153</td> <td>178</td> </tr> <tr> <th>Units</th> <td>40</td> <td>42</td> <td>53</td> <td>35</td> <td>56</td> <td>39</td> <td>48</td> <td>30</td> <td>37</td> <td>40</td> </tr> </table> a. Develop the regression equation for the cost accountants. b. Predict overhead when 50 units are produced. c. Calculate the error of estimate. d. Write python code to plot true line	Overheads	191	170	272	155	280	173	234	116	153	178	Units	40	42	53	35	56	39	48	30	37	40			
Overheads	191	170	272	155	280	173	234	116	153	178																	
Units	40	42	53	35	56	39	48	30	37	40																	

COURSE: BCA
SUBJECT: Intro. to Artificial Intelligence
PAPER CODE: BCAT-214

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A		10 Marks	Marks
Q. No.	CO	Question	5x2
1.		Attempt the following Questions (Do any 5):-	
	CO1	i. What constitutes a state space, and how can it be illustrated using the example of a Tic-Tac-Toe game?	
	CO2	ii. What is Heuristic Search, and how does it aid in navigating search spaces?	
	CO2	iii. Explain the notion of constraint satisfaction and its significance in problem-solving.	
	CO1	iv. Define Artificial Intelligence (AI) and enumerate its various real-world applications.	
	CO1	v. What sets apart informed search algorithms from uninformed ones?	
	CO1	vi. How do Depth-First Search (DFS) and Breadth-First Search (BFS) algorithms differ in their approach to traversing graphs?	
	CO1	vii. What challenges arise when formulating a search problem?	

PART - B		2*10 = 20 Marks	Marks
Q. No.	CO	Question	(2.5 x 4 = 10)
2	CO2	Explain the following search algorithms with example: (a) Best first search (b) AO* (c) MEA search algo (d) Hill climbing	5
3	CO3	(a) What is meant by the term "agent," and what are your thoughts on the assertion that autonomy is a controlling concept in agent design? (b) Examine the different issues encountered in the Hill Climbing Algorithm and propose methods for addressing them.	5
4	CO1	(a) What sets apart Predicate Logic from Propositional Logic? (b) How do forward and backward search approaches differ, and can you provide an illustrative example?	5

COURSE: BCA
SUBJECT: Network Security
PAPER CODE: 216

SEMESTER: IV
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A		10 Marks	Marks
Q. No.	CO	Question	10
1.	1 1 1 4 6	Differentiate between the following a. Active & Passive Attacks b. IDS & IPS c. Symmetric & Asymmetric key cryptography d. MD5 & SHA e. TLS & SSL	

PART - B		2*10 = 20 Marks	Marks
Q. No.	CO	Question	10
2	2, 3	Define Firewall? What are the design principles of firewalls? Explain the various types of firewalls.	10
3	1	Explain network security services & security mechanisms. Also draw and describe network security model.	10
4	4	What is PKI? Describe its architecture.	10
5	5	Discuss RSA Algorithm? Assume p=3, q=11, e=7 calculate d and cipher text for the message is B using RSA algorithm.	10

COURSE: BCA
SUBJECT: Web Development with Python and Django
PAPER CODE: BCAT 218

SEMESTER: IV
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A		10 Marks	Marks
Q. No.	CO	Question	5
1.		Attempt all parts	5
	CO1	i) Differentiate between MVC and MVT design patterns with diagrams.	
	CO2	ii) Create a simple template and render it in the corresponding view by taking a suitable example. Show all necessary files.	

PART - B		2*10 = 20 Marks	Marks
Q. No.	CO	Question	10
2	CO4	Design a form in Django with at least five different form fields and proper validations. Explain all steps and fields in detail. Also, demonstrate how to get the form data and display it in the appropriate template.	5*2=10
3	CO4	i) Explain how you can convert the list of choices using widgets to: a. A list of radio buttons b. A list of checkboxes	
		ii) What is a view? Explain how to create a dynamic view with a suitable example. Show all necessary files.	
4	CO1	i) Explain the following: a. HTML Auto Escaping and its template tags b. Loose Coupling in Django	
	CO2, CO1	ii) What is template inheritance? Explain template inheritance tags with suitable example.	5*2=10
	CO2		

COURSE: BCA

SUBJECT: Digital Marketing

PAPER CODE: BCA 222

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST - APRIL - 2024

SEMESTER: IV

M.MARKS: 30

TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A

Q. No.	CO	Question	Marks
1.	CO 1	What is the difference between inbound and outbound marketing?	5
2	CO 2	Explain various channels in Digital Marketing.	5

PART - B

Q. No.	CO	Question	Marks
3	CO2	Explain POEM framework in detail.	10
4	CO3	Nutella – Incredible content that makes you salivate Take your #breakfast bread pudding to the next level with #Nutella!  Each post makes you want to eat Nutella. There are a lot of people who take photos of their food before they eat it. Nutella does the same and it works. Nutella isn't afraid to be fun and creative with different ingredients. Nutella is just a chocolate spread yet they manage to have fun with it. What lessons can you learn from this in terms of: - Social Media Channels - Influencers / Bloggers - Followers	10
5	CO4	Few mainstream brands are as fascinating SEO-wise as Wise. Formerly known as Transfer Wise, this fintech company offering multi-currency accounts and quick international money transfers with fair fees currently gets an estimated 6.4 million organic visits per month:  That's especially impressive considering they moved domains in early 2021—a migration so well-executed that it made waves in the SEO community. Add to this that they have a custom CMS to scale their SEO efforts. Discuss the above case study in terms of your suggestions on: - Keyword Search - Organic and Nonorganic SEO - Content Strategy	10

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST – APRIL - 2024

COURSE: BCA
 SUBJECT: Intro to Management & Entrepreneurship Development
 PAPER CODE: BCA 206

SEMESTER: 4
 M.MARKS: 30
 TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B

PART – A

Q. No.	CO	Question	10 Marks
1.	CO1	What is the role of entrepreneurship in the economic development of the country?	10

PART – B

2*10 = 20 Marks

Q. No.	CO	Question	Marks
2	CO2,CO3	Define the meaning of an entrepreneur. Also, explain the general characteristics and personality traits of an entrepreneur and name any 5 agencies involved in entrepreneurship development in India.	10
3	CO2,CO3	Define creativity. Explain why it's necessary for entrepreneurs to be creative. Provide two examples of creative entrepreneurship from the industry.	10
4	CO2,CO4	Define Innovation. Also, explain the principles of identification of opportunities for problem-solving with innovation. Give 2 examples of innovation from the industry.	10

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST – APRIL - 2024

BCA

SOFTWARE ENGINEERING

COURSE CODE: BCA 204

SEMESTER: IV

M.MARKS: 30

TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B

10 Marks

PART – A

Q. No.	CO	Question	Marks
1.	CO3 CO5 & CO6 CO1 CO5	a) What is requirement engineering? Explain various requirement elicitation techniques. b) What is risk? What are various risk management activities? c) Explain the Software Crisis with a suitable example. d) Explain Halstead Software Science Measure for any five measures.	2.5*4=10

2*10 = 20 Marks

PART – B

PART – B

Q. No.	CO	Question	Marks																				
2	CO5	a. Consider a project with 10 low external inputs, 13 high external outputs, 15 low internal logical files, 17 High external interface files, 12 average external enquiries. In addition to above, system requires Essential backup and recovery, significant data communication capability; Designed code may be moderately reusable. Other complexity adjustment factors and weighting factors are treated as average. What are the unadjusted and adjusted function point counts? b. The project size of 400 KLOC is to be developed. The project schedule is very tight. Calculate the effort, development time, average staff size, and productivity for the project using the COCOMO Model. <table> <tr> <th>Project</th> <th>a_b</th> <th>b_b</th> <th>c_b</th> <th>d_b</th> </tr> <tr> <td>Organic</td> <td>2.4</td> <td>1.05</td> <td>2.5</td> <td>0.38</td> </tr> <tr> <td>Semidetached</td> <td>3.0</td> <td>1.12</td> <td>2.5</td> <td>0.35</td> </tr> <tr> <td>Embedded</td> <td>3.6</td> <td>1.20</td> <td>2.5</td> <td>0.32</td> </tr> </table>	Project	a _b	b _b	c _b	d _b	Organic	2.4	1.05	2.5	0.38	Semidetached	3.0	1.12	2.5	0.35	Embedded	3.6	1.20	2.5	0.32	5*2=10
Project	a _b	b _b	c _b	d _b																			
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Embedded	3.6	1.20	2.5	0.32																			
3	CO2	Consider the problem of an Online Shopping System and design the following: a. Use case diagram b. Context Diagram c. 1-level DFD d. Class Diagram	10																				
	CO4	Explain Agile Model in detail. What are the advantages and disadvantages of Agile Model? Compare and Contrast with Waterfall, Prototype and Spiral SDLC Models.	10																				